

Trainings ▾

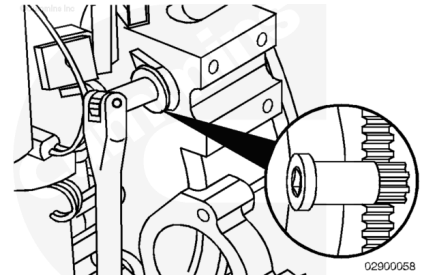
Rotation Check

If the starter solenoid is making a sound but the engine is **not** rotating, turn the keyswitch to the OFF position, and attempt to bar the crankshaft in both directions.

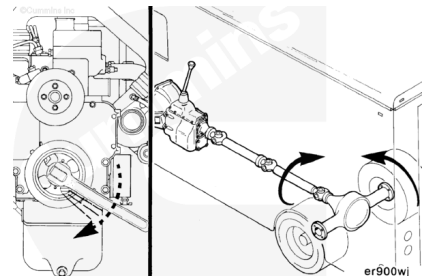
Bar the engine with the barring tool, Cummins® Part Number 3824591.

If the crankshaft will bar over, attempt to start the engine. If the starter motor cranks the engine, check the starter motor pinion gear and flywheel ring gear for damage.

If damage to the starter motor pinion gear and/or flywheel ring gear is found when replacing the components, make sure to measure the distance from the starting motor mounting flange to the forward face of the front side of the flywheel ring gear. Follow the measure step of this procedure.



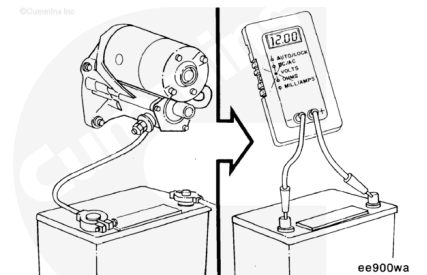
If the crankshaft does **not** rotate or requires more than the normal effort to bar, check for an internal malfunction or a problem with the drive unit and/or accessories.



If the engine cranking speed is too slow/will **not** crank at all, and the engine rotates freely:

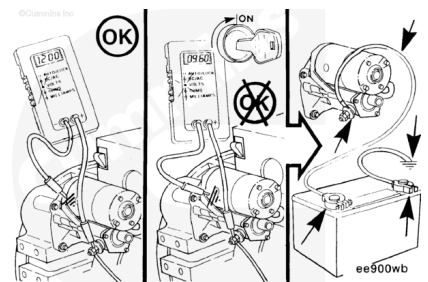
Make sure the wiring connections are clean, tight and **not** damaged.

Check the battery voltage. Refer to Procedure 013-007 in Section 13 (</qs3/pubsys2/xml/en/procedures/40/40-013-007.html>).



Check the voltage at the starting motor during cranking. If the voltage drops more than 2.4-VDC on a 12 volt system and 4.8 volt on a 24-VDC system, check that all connections are clean and tight.

If the cables are correct and the voltage drop exceeds the limit, replace the starting motor.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

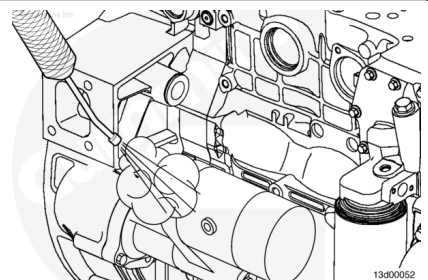


- Disconnect the ground cable from the battery terminal. See equipment manufacturer service information.
- Identify each wire with a tag indicating location on starting motor.
- Remove the electrical connections from the starting motor.

Remove

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

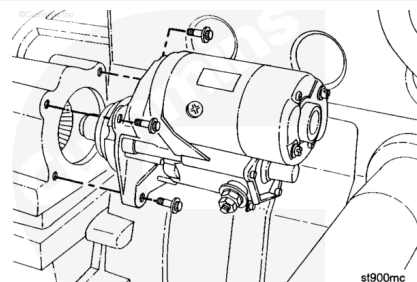
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Prior to removing the starter, use steam to clean the area around the starting motor to prevent debris from entering the flywheel housing.

Dry with compressed air.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

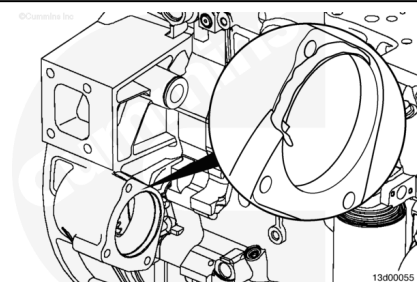


Remove the three capscrews and the starting motor.

Note : If equipped with a starting motor spacer, remove the spacer and clean all surfaces between the starting motor, starting motor spacer, and flywheel housing with a wire brush.

Clean and Inspect for Reuse

For engines that use wet flywheel housings, clean any left over sealant from the starting motor mounting flange on both the flywheel housing and starting motor. Make sure these surfaces are clean of oil and debris.



Inspect the starter motor pinion gear and/or flywheel ring gear for chipping or uneven wear.

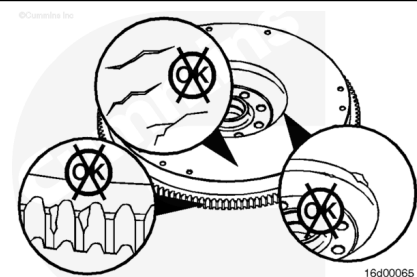
Note : If the starter motor pinion gear and/or flywheel ring gear teeth are damaged, they must be replaced.

Use the following procedure, if equipped with a flywheel. Refer to Procedure 016-005 in Section 16.

(</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>),

Use the following procedure, if equipped with a flexplate. Refer to Procedure 016-004 in Section 16

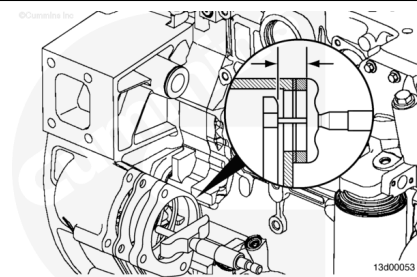
(</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>).



Measure

Using an inside micrometer or a vernier caliper, measure the distance from the starting motor mounting flange to the forward face of the front side of the flywheel ring gear.

Note : Include any spacers previously removed when completing the measurement.



Starting Motor Spacing

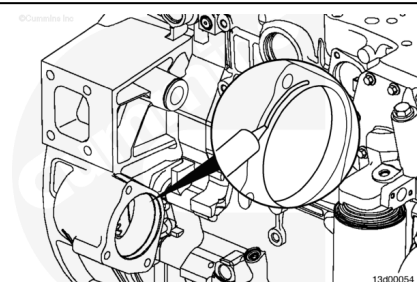
mm		in
49.28	MIN	1.94
52.32	MAX	2.06

Add or remove spacers as necessary to achieve the correct starting motor spacing.

Install

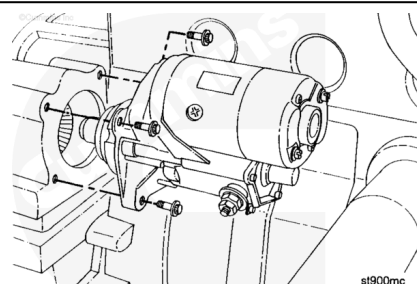
For engines with wet flywheel housings, apply a 1.5 to 2.0 mm [0.06 to 0.09 in] wide bead of sealant, Part Number 3164067, to the flywheel housing starting motor mounting flange.

Note : If a starting motor spacer is required, make sure to apply sealant to the side of the spacer that contacts the starting motor.



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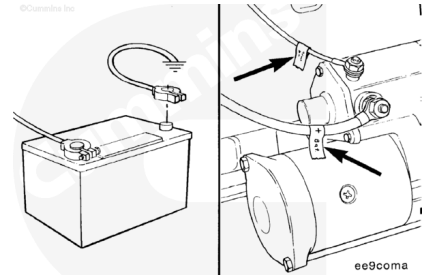
Install the three capscrews, the starting motor, and starting motor spacer, if required.

Torque Value: 43 n•m [32 ft-lb]

Cummins® Branded Starters

⚠ CAUTION ⚠

Do not overtighten the electrical connections. Starter damage can result.



Note : Use the location tags to help identify where each wire connection goes.

Connect the electrical connections to the starter motor.

Torque Value:

M5

1.4 n•m [35 in-lb]

Torque Value:

M10

1.21 n•m [185 in-lb]

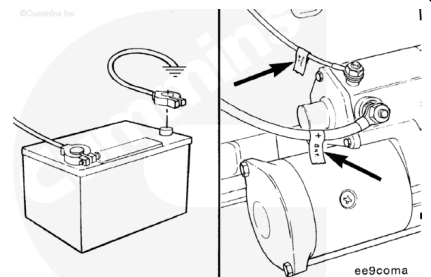
Install the jump start protection (JSP) cover and cover nut on the M terminal post.

Note : The JSP cover nut is the third nut on the M terminal, M5 terminal size. Failure to observe the proper torque specification can result in loss of conductivity to the M lead and result in a no crank condition for the starter and engine.

Non-Cummins® Branded Starters

⚠ CAUTION ⚠

Do not overtighten the electrical connections. Starter damage can result.



Note : Use the location tags to help identify where each wire connection goes.

Install the starter motor electrical connections.

For Non-Cummins® branded starters, refer to the OEM manual for torque specifications.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the ground cable to the battery terminal.

